Applied Chemistry — Last-Minute Quiz Sheet

Order: 3A Electrical & Magnetic → 3B Nano, Quantum dots, Coolants. Only what is likely to be asked.

All formulas on one line each

$V = IR$ · $R = \rho L/A$ · $\rho_T = \rho_0(1 + \alpha\Delta T)$ · $H_c(T) = H_c(0)[1 - (T/T_c)^2]$

$\chi = M/H$ · $\mu = B/H$ · $\mu_r = 1 + \chi$ · $\mu_0 = 4\pi\times10^{-7}$ H/m · $\mu_{eff} = \sqrt{(n(n+2))}\ \mu_B$ · $\mu_B = 9.27\times10^{-24}$ J/T

Para: $\chi = C/T$ (Curie) · Ferro above T_C: $\chi = C/(T - T_C)$ (Curie–Weiss) · de Broglie $\lambda = h/mv$

Module 3A — Electrical materials

- Ohm's law holds only at **constant temperature** and for ohmic materials. Diodes and transistors are **non-ohmic**.
- Resistance increases with length and decreases with area. Resistivity ρ (Ω·m) depends only on the **material and temperature**, not the shape.

Type	ρ (Ω·m)	Band gap	α
Conductor	10^{-8}–10^{-6}	≈ 0 (bands overlap)	+ve
Semiconductor	10^{-5}–10^{6}	~0.7–1.5 eV	−ve
Insulator	10^{8}–10^{16}	> 5 eV	−ve

- In metals ρ rises with temperature because lattice vibration scatters electrons. In semiconductors ρ falls with temperature because more electrons reach the conduction band.
- **Nichrome / constantan** have very low α, so they are used for heating coils and standard resistors. Semiconductors with negative α are used as **thermistors**.
- Factors that raise ρ in metals: impurities, strain, crystal defects. **Annealing** removes defects and lowers ρ.
- Band gaps: diamond ~5.5 eV, glass ~9 eV.

Cu vs Al

	Cu	Al
Config	[Ar]$3d^{10}4s^1$	[Ne]$3s^23p^1$
ρ	1.7×10^{-8}	2.7×10^{-8}
Density	8.9 g/cm³	2.7 g/cm³
Use	house wiring, PCB	overhead lines, bus bars

- Silver has the lowest ρ (1.6×10^{-8}). Copper is second.
- Al has ~61% of copper's conductivity but is much lighter. **ACSR** cable = aluminium conductor + steel core for strength. Al is protected by its Al_2O_3 film.
- **Hard-drawn Cu** is cold-worked: stronger, lower conductivity, used for overhead lines. **Annealed Cu** is soft and ductile with higher conductivity, used for house wiring.

Semiconductors

- Intrinsic semiconductors are pure Si or Ge. They act as insulators at 0 K. Current is carried by electrons **and** holes.
- Doping level is about 1 impurity atom in 10^8.

	n-type	p-type
Dopant	Group V: P, As, Sb, Bi	Group III: B, Al, Ga, In
Called	Donor	Acceptor
Majority carrier	Electrons	Holes
Impurity ion	+ve	−ve

- Compound semiconductors: **GaAs** (LEDs, laser diodes), **InP** (optical and high-speed communication).

Superconductors

- Below T_c the resistance is **zero**, so a current can persist indefinitely.
- **Onnes, 1911**: mercury at **4.2 K** (Nobel 1913, he also liquefied helium).
- **Meissner effect (1933, with Ochsenfeld)**: magnetic flux is expelled, so the material is a **perfect diamagnet, χ = −1**. This is what separates a superconductor from a merely perfect conductor.
- **Bednorz & Müller, 1986**: high-temperature superconductors (Nobel 1987).
- **YBCO ($YBa_2Cu_3O_7$)**: $T_c \approx$ **92 K**, above liquid N_2 (**77 K**). It is Type II, a perovskite-like cuprate whose superconductivity sits in the **CuO_2 planes**. H_{c2} > 100 T, J_c ~10^6 A/cm².

Type I (soft)	Type II (hard)
Complete Meissner effect, one low H_c, sharp transition	Partial flux entry (vortices), two fields H_{c1} and H_{c2}, high current
Hg, Pb, Al	NbTi, Nb_3Sn, YBCO

T_c values: Hg 4.2 · Pb 7.2 · Nb 9.2 · NbTi 10 · Nb_3Sn 18 · YBCO 92 · BSCCO ~110 K

Uses: MRI, NMR, Maglev (SCMaglev > 600 km/h), **SQUID** (loop with 2 Josephson junctions, very sensitive magnetometer, brain mapping by MEG), qubits, LHC, ITER, SMES, fault-current limiters.
Josephson junction = 2 superconductors separated by a thin insulator.

Module 3A — Magnetic materials

- Magnetism comes from the electron's **orbital motion + spin**. **Unpaired electrons** give a net moment. $g_S = 2$.
- Formula check: n = 1 gives μ = 1.73 μ_B, n = 2 gives 2.83, n = 3 gives 3.87, n = 4 gives 4.90, n = 5 gives 5.92.

Type	χ	μ_r	Key behaviour	Examples
Dia	small, −ve	< 1	All electrons paired, weakly repelled, **independent of T**	Bi, Cu, Ag, Au, water, graphite
Para	small, +ve	> 1	Unpaired spins randomly oriented, weakly attracted, $\chi = C/T$	Al, Pt, Na, O_2
Ferro	≫ 1	10^2–10^5	Domains, hysteresis, permanent magnet; above T_C becomes paramagnetic	Fe, Co, Ni, Gd
Antiferro	small, +ve	≈ 1	Equal antiparallel spins, net ≈ 0, **χ peaks at the Néel temp T_N**	MnO, FeO, NiO, CoO
Ferri	large, +ve	10^2–10^4	Unequal antiparallel spins, so a net moment remains	Fe_3O_4, ferrites

- Curie temperatures: Fe 1043 K, Co 1404 K, Ni 631 K, Gd 293 K. Néel temperatures: Cr 308 K, Mn 100 K.
- Storage: **HDD** stores bits as domain orientation (0/1). Tapes and floppies work the same way. **Magneto-optical** drives use a laser to heat a spot **above the Curie temperature** so it can be rewritten. **MRAM** is non-volatile.
- Soft magnets (high μ, low loss) are used in transformer cores. Hard magnets (NdFeB, SmCo) are permanent magnets. Mu-metal is used for magnetic shielding.

Module 3B — Nanomaterials

- A nanomaterial has **at least one dimension of 1–100 nm**. 1 nm = 10^{-9} m, about 10 atoms side by side. A human hair is 50,000–100,000 nm across.
- **0D**: all three dimensions are nanoscale (nanoparticles, quantum dots). **1D**: nanowires, nanorods, nanotubes. **2D**: thin films, nanolayers. **3D**: bulk built from nano units.
- Two reasons nano behaves differently: **(1) a high surface-area-to-volume ratio**, which makes it a better catalyst, and **(2) quantum confinement**, which gives discrete levels and a larger band gap.
- Cube example: one 1 cm cube has 6 cm^2 of surface. Cut into 8 cubes it has 12 cm^2. Cut into 27 cubes it has 18 cm^2. The volume stays the same.
- Confinement starts when particle size is comparable to the electron's **de Broglie wavelength**.
- Uses: TiO_2 (sunscreen, self-cleaning glass), nano-silica (dental fillers), nano-clay (tennis balls), MnO_2 (batteries).

Top-down vs bottom-up

Top-down (break bulk down)	Bottom-up (build from atoms)
Mechanical (ball) milling, etching, laser ablation, sputtering, electro-explosion	Sol-gel, CVD, chemical reduction, green synthesis, laser pyrolysis, supercritical fluid, spinning
Long-range order	Short-range order, uniform (monosized) particles

Ball milling (top-down)

- Also called mechanical alloying or mechanochemical synthesis.
- ~50 μm powder is milled with hardened steel or **WC-coated balls**, at a ball:powder mass ratio of 5:10.
- Mill types: tumbler, attrition, shaker (SPEX 8000), vibratory, planetary.
- Products: nanocrystalline, amorphous and quasicrystalline phases.

Bulk vs nano gold

Bulk Au	Nano Au
Yellow, lustrous	Many colours depending on size and shape (due to **surface plasmon resonance**)
Metal, conductor	Behaves like a semiconductor (slide: Eg ≈ 3.4 eV)
Melting point ~1080 °C	Lower melting point (~940 °C)
Inert	Very good catalyst
Malleable, ductile	—

Gold particles in glass: **25 nm reflects red**, **50 nm green**, **100 nm orange**.
Silver: 100 nm sphere is yellow, 40 nm sphere blue, 100 nm prism red.

- Advantage: **high production rate**, cheap.
- Disadvantages: **contamination** (Fe from the tools, O_2/N_2 from air, so a glove box is needed), local heating to 100–200 °C, and runs over 30 h push contamination above 10%.

Sol-gel (bottom-up)

- Precursor is a **metal alkoxide**, e.g. $Si(OR)_4$ (TEOS).
- Steps: **hydrolysis → condensation → aging → drying → calcination**.
- Hydrolysis: M–OR + H_2O → M–OH + ROH.
- Condensation forms **oxo (M–O–M)** or **hydroxo (M–OH–M)** bridges. For silicon these are **siloxane (Si–O–Si)** bridges, releasing H_2O or ROH.
- **Xerogel** is dried by normal evaporation. **Aerogel** is dried by **supercritical** extraction.
- Advantages: low temperature, monosized particles. Disadvantages: **slow**, agglomeration, leftover reactants.

Module 3B — Quantum dots

- QDs are **semiconductor** nanocrystals of **2–10 nm** (10–50 atoms across, 100–1000 electrons). Examples: **CdSe, CdS, InP**.
- **Smaller dot → larger band gap → higher-energy emission (blue/violet)**, around 2–3 nm.
- **Larger dot → smaller gap → red/orange emission (red shift)**, around 5–6 nm.
- Emission colour is set by size, so it is tunable, narrow and a single colour.
- Properties: high extinction coefficient, high electron mobility, solution-processable.
- Synthesis: high-temperature hot injection (above 200 °C, N_2), **hydrothermal** (Teflon-lined steel **autoclave**), microwave (e.g. carbon dots from starch), ultrasonication.
- Uses: **QD-OLED displays**, solar cells, bio-imaging, drug delivery, LEDs, lasers, metal-ion sensing.

Module 3B — Liquid coolants

- A good coolant has **high thermal conductivity, specific heat and latent heat, with low viscosity**.
- Chips can produce heat fluxes up to **1000 W/cm²**.
- Electronics need **low electrical conductivity**. Water and glycol-water conduct, so non-aqueous dielectric coolants are preferred.
- A **flash point below 93 °C** counts as flammable (ethanol, methanol, benzene). PFC and HFE are non-flammable.
- Toxicity: fluorocarbons and HFEs are less toxic than ethylene glycol.
- Environment: **PFC has high GWP and zero ODP**. **HFE has low GWP and zero ODP**, so it is the greener choice.
- Cu and Al are the preferred wetted metals: high thermal conductivity and compatible with most coolants.
- **Water**: high cp and k, cheap, low risk. Downside: it freezes.
- **Deionised water**: less corrosive than tap water, but still needs inhibitors: **phosphates, tolyltriazole, 2-ethylhexanoic acid**.
- **Water-glycol** (ethylene or propylene glycol) has a lower freezing point.
- **NaCl / $CaCl_2$ brines** have a low freezing point but are **highly corrosive**.
- **K formate/acetate**: freezing point about **−70 °C**, less corrosive. Used in the food and pharma industries.
- **Dielectric coolants** are used for data-centre immersion and direct-to-chip cooling:
  - Aromatics: toluene, diethylbenzene.
  - Aliphatics: paraffins, PAO.
  - Fluorocarbons: HFE-7100, FC-72.
  - Silicones: PDMS, Syltherm.

Solved numericals (exact slide problems)

1. Spin-only moment. Fe^{3+} (d^5, n = 5): μ = √35 = **5.92 μ_B**.
Cu^{2+} (d^9, n = 1): μ = **1.73 μ_B**.

2. Resistance. Doubling L doubles R. Doubling the diameter makes A four times bigger, so R falls to **R/4**.

3. $H_c(T)$. At T = T_c/2: $H_c = H_c(0)(1 - ¼)$ = **0.75 $H_c(0)$**.

Practice MCQs

1. As temperature rises, the resistivity of a semiconductor:
   a) Increases b) Decreases c) Stays constant d) Becomes zero

2. Majority carriers in P-doped Si:
   a) Holes b) Electrons c) Ions d) Protons

3. Magnetic susceptibility of a superconductor:
   a) 0 b) +1 c) −1 d) ∞

4. T_c of YBCO is about:
   a) 4.2 K b) 18 K c) 77 K d) 92 K

5. Which is a Type I superconductor?
   a) Nb_3Sn b) YBCO c) Pb d) NbTi

6. Magnetic susceptibility that is independent of temperature:
   a) Para b) Dia c) Ferro d) Antiferro

7. Susceptibility peaks at the Néel temperature in:
   a) Ferro b) Ferri c) Antiferro d) Para

8. Fe_3O_4 is:
   a) Ferromagnetic b) Ferrimagnetic c) Diamagnetic d) Antiferromagnetic

9. Rewriting data in magneto-optical storage uses heating above the:
   a) Néel temp b) Curie temp c) Melting point d) Tc of a superconductor

10. The sol-gel method is:
    a) Top-down b) Bottom-up c) Both d) Neither

11. The precursor in sol-gel synthesis is usually a:
    a) Metal halide b) Metal alkoxide c) Metal carbide d) Pure metal

12. An aerogel is dried by:
    a) Evaporation b) Supercritical extraction c) Calcination d) Freezing

13. The main drawback of ball milling is:
    a) Slow rate b) Contamination c) High cost d) Monosized particles

14. Smaller quantum dots emit:
    a) Red b) Orange c) Blue d) Infrared

15. Colour of 25 nm gold nanoparticles in glass:
    a) Yellow b) Red c) Green d) Orange

16. A coolant with low GWP and zero ODP:
    a) PFC b) HFE c) Ethanol d) Brine

17. Freezing point of potassium formate/acetate coolant is about:
    a) 0 °C b) −20 °C c) −70 °C d) −196 °C

18. Spin-only magnetic moment for 3 unpaired electrons:
    a) 1.73 b) 2.83 c) 3.87 d) 4.90 μ_B

19. A conductor is used for overhead lines because of its low density:
    a) Cu b) Ag c) Al d) Au

Answers: 1 b · 2 b · 3 c · 4 d · 5 c · 6 b · 7 c · 8 b · 9 b · 10 b · 11 b · 12 b · 13 b · 14 c · 15 b · 16 b · 17 c · 18 c · 19 c